NAME

mbm_makeimagelist − Generate an MB-System stereo imagelist file from a MBARI Low Altitude Survey System (LASS) camera data directory.

VERSION

Version 5.0

SYNOPSIS

mbm_makeimagelist [**−−help −−check−bayer −−imdir=***path2images* **−−imformat=***image_format* **−−outfile=***outfname* **−−gpsfile=***gpsfname* **−−nogpsfile −−nogains**]

DESCRIPTION

Mbm_makeimagelist is a Python macro that generates an MB-System stereo imagelist file from the directory structure and filenames used by the MBARI Low Altitude Survey System (LASS) stereo seafloor photography system. It expects to find left camera images in a subdirectory named *PROSILICA_L* and right camera images in a subdirectory named *PROSILICA_R* of the input image directory, with each image file's basename being an integer image acquisition timestamp expressed in microseconds since the Unix epoch.

The left and right image time lists are matched into stereo pairs. By default this matching also uses a separate GPS trigger time log (a CSV file whose first column is a microsecond epoch trigger time for each stereo capture, by default named *logfile_gpstime_cam_sync.csv* one directory above the image directory); each left and right image is paired with the closest GPS trigger time, and left or right images with no corresponding trigger within a fixed time tolerance are recorded as unmatched. If **−−nogpsfile** is given, the left and right image time lists are instead matched directly against each other using the same time tolerance, without reference to any GPS trigger log.

Unless **−−nogains** is given, **mbm_makeimagelist** also opens every input image with the Python Imaging Library and reads its embedded TIFF *ImageDescription* tag (tag 270), extracting the first two numeric values found in that tag's text as the camera's gain and exposure settings at the time the image was captured. If **−−check−bayer** is given, the word "color" in each image's filename is first replaced with "bayer" before the file is opened to read this tag, on the premise that the original raw Bayer-pattern image (rather than a subsequently color-processed copy) is more likely to carry the camera gain/exposure metadata; this substitution only affects which file is opened to read the gain and exposure values; it does not change which image files are listed in the output imagelist.

The result is written as a single MB-System imagelist file in the *.mb−2* format, with one line per stereo pair (or single, unmatched image) giving the left image path (or **NULL** if there is no left image), the right image path (or **NULL**), the paired time (twice), and the left and right camera gain and exposure values (zero if **−−nogains** was used or no metadata could be read). The first five lines of the data are additionally prefixed with a "##" comment marker in the output file.

MB-SYSTEM AUTHORSHIP

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OPTIONS

−h, −−help
This "help" flag causes the program to print out a description of its command line options and then exit immediately.

−b, −−check−bayer
Reads camera gain and exposure settings from the corresponding raw Bayer pattern image file rather than from the (possibly color-processed) image file actually being listed, by substituting "bayer" for "color" in the filename used only for that metadata lookup. Has no effect if **−−nogains** is also given.

−d, −−imdir=*path2images*
Sets the path to the directory containing the *PROSILICA_L* and *PROSILICA_R* image subdirectories to be listed. The default is the current directory (*./*).

−f, −−imformat=*image_format*
Sets the image file suffix (for example *tif* or *jpg*) searched for in the *PROSILICA_L* and *PROSILICA_R* subdirectories. The default is *tif*.

−o, −−outfile=*outfname*
Sets the name of the output imagelist file to be written, without the *.mb−2* suffix, which is appended automatically. If *outfname* includes a directory path, the output file is written to that directory; otherwise it is written to the image directory set by **−−imdir**. The default output filename is *imagelist*, producing *imagelist.mb−2*.

−g, −−gpsfile=*gpsfname*
Sets the filename of the GPS trigger time CSV log used to match left and right images into stereo pairs. The default is *./../logfile_gpstime_cam_sync.csv* relative to the image directory (that is, one directory above **−−imdir**). Has no effect if **−−nogpsfile** is given.

−x, −−nogpsfile
Matches left and right camera images into stereo pairs directly by comparing their timestamps, without reading or requiring a GPS trigger time log file.

−y, −−nogains
Skips opening the image files to read embedded camera gain and exposure metadata; the gain and exposure values written to the output imagelist are all zero. This significantly speeds up processing of large image sets when gain/exposure correction is not needed.

SEE ALSO

mbsystem(1), **mbimagelist**(1), **mbgetphotocorrection**(1), **mbphotomosaic**(1), **mbphotogrammetry**(1)

BUGS

Maybe. Maybe not.